

# BIMBINGAN TUGAS AKHIR #5

Jumat, 9 Oktober 2026

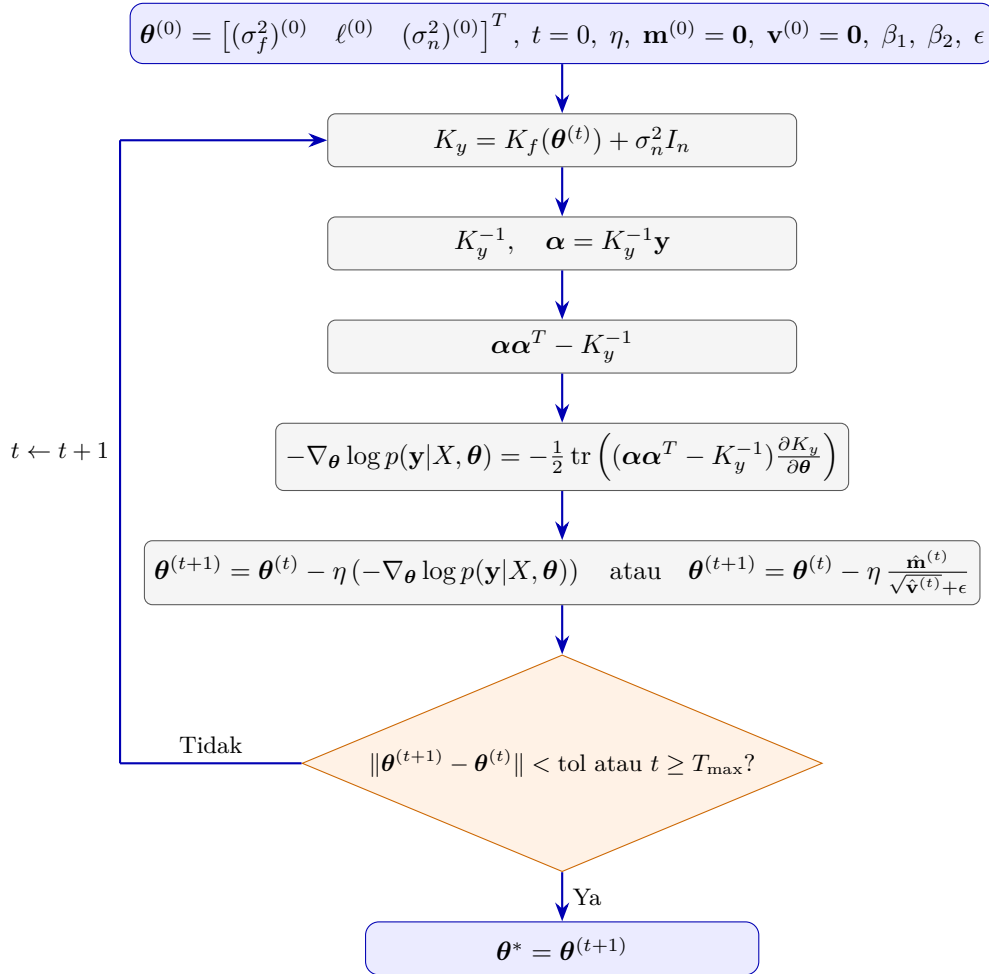
## 1 Contoh Perhitungan Optimasi Log Marginal Likelihood ( $2 \times 2$ )

(Rasmussen & Williams, 2006, Bab 5.4.1, hlm. 113–114, Persamaan (5.9))

$$\frac{\partial \log p(\mathbf{y}|X, \boldsymbol{\theta})}{\partial \theta_j} = \frac{1}{2} \text{tr} \left( (\boldsymbol{\alpha} \boldsymbol{\alpha}^T - K_y^{-1}) \frac{\partial K_y}{\partial \theta_j} \right), \quad \text{dengan } \boldsymbol{\alpha} = K_y^{-1} \mathbf{y} \quad (1)$$

Karena algoritma optimasi (seperti *Gradient Descent* maupun *Adam*) bertujuan untuk meminimumkan suatu fungsi objektif sedangkan tujuan kita adalah memaksimalkan *log marginal likelihood*, maka fungsi objektif yang diminimumkan dikalikan  $-1$  ( $-\log p(\mathbf{y}|X, \boldsymbol{\theta})$ ):

$$-\frac{\partial \log p(\mathbf{y}|X, \boldsymbol{\theta})}{\partial \theta_j} = -\frac{1}{2} \text{tr} \left( (\boldsymbol{\alpha} \boldsymbol{\alpha}^T - K_y^{-1}) \frac{\partial K_y}{\partial \theta_j} \right) \quad (2)$$



Gambar 1: Diagram Iterasi Optimasi Hyperparameter.

## 1.1 Gradient Descent

### 1. Data Observasi & Parameter Awal

- Data Input:  $X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ , Target:  $\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$
- Jarak:  $r_{12} = |x_1 - x_2| = 1 \implies r_{12}^2 = (x_1 - x_2)^2 = 1$
- Kernel RBF:  $k(x_p, x_q) = \sigma_f^2 \exp\left(-\frac{r_{pq}^2}{2\ell^2}\right)$
- Inisialisasi:  $\boldsymbol{\theta}^{(0)} = \begin{bmatrix} (\sigma_f^2)^{(0)} \\ \ell^{(0)} \\ (\sigma_n^2)^{(0)} \end{bmatrix} = \begin{bmatrix} 1,0 \\ 1,0 \\ 0,1 \end{bmatrix}$ , Learning Rate:  $\eta = 0,1$

### 2. Matriks Kovariansi $K_y$

$$k_{11} = k_{22} = \sigma_f^2 = 1,0 \quad (3)$$

$$k_{12} = k_{21} = 1,0 \cdot \exp\left(-\frac{1}{2(1,0)^2}\right) = e^{-0,5} \approx 0,6065 \quad (4)$$

$$K_y = \begin{bmatrix} \sigma_f^2 + \sigma_n^2 & k_{12} \\ k_{21} & \sigma_f^2 + \sigma_n^2 \end{bmatrix} = \begin{bmatrix} 1,0 + 0,1 & 0,6065 \\ 0,6065 & 1,0 + 0,1 \end{bmatrix} = \begin{bmatrix} 1,1000 & 0,6065 \\ 0,6065 & 1,1000 \end{bmatrix} \quad (5)$$

### 3. Determinan, Invers $K_y^{-1}$ , dan Vektor $\boldsymbol{\alpha}$

$$|K_y| = (1,1000)^2 - (0,6065)^2 = 1,2100 - 0,3678 = 0,8422 \quad (6)$$

$$K_y^{-1} = \frac{1}{0,8422} \begin{bmatrix} 1,1000 & -0,6065 \\ -0,6065 & 1,1000 \end{bmatrix} = \begin{bmatrix} 1,3062 & -0,7202 \\ -0,7202 & 1,3062 \end{bmatrix} \quad (7)$$

$$\boldsymbol{\alpha} = K_y^{-1} \mathbf{y} = \begin{bmatrix} 1,3062 & -0,7202 \\ -0,7202 & 1,3062 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} (1,3062)(1) + (-0,7202)(2) \\ (-0,7202)(1) + (1,3062)(2) \end{bmatrix} = \begin{bmatrix} -0,1342 \\ 1,8922 \end{bmatrix} \quad (8)$$

### 4. Matriks $(\boldsymbol{\alpha}\boldsymbol{\alpha}^T - K_y^{-1})$

$$\begin{aligned} \boldsymbol{\alpha}\boldsymbol{\alpha}^T &= \begin{bmatrix} -0,1342 \\ 1,8922 \end{bmatrix} \begin{bmatrix} -0,1342 & 1,8922 \end{bmatrix} \\ &= \begin{bmatrix} (-0,1342)^2 & (-0,1342)(1,8922) \\ (1,8922)(-0,1342) & (1,8922)^2 \end{bmatrix} = \begin{bmatrix} 0,0180 & -0,2540 \\ -0,2540 & 3,5805 \end{bmatrix} \end{aligned} \quad (9)$$

$$\boldsymbol{\alpha}\boldsymbol{\alpha}^T - K_y^{-1} = \begin{bmatrix} 0,0180 & -0,2540 \\ -0,2540 & 3,5805 \end{bmatrix} - \begin{bmatrix} 1,3062 & -0,7202 \\ -0,7202 & 1,3062 \end{bmatrix} = \begin{bmatrix} -1,2882 & 0,4662 \\ 0,4662 & 2,2743 \end{bmatrix} \quad (10)$$

## 5. Perhitungan Gradien

a. Terhadap  $\sigma_f^2$ :

$$\frac{\partial K_y}{\partial \sigma_f^2} = \begin{bmatrix} 1 & 0,6065 \\ 0,6065 & 1 \end{bmatrix} \quad (11)$$

$$(\alpha\alpha^T - K_y^{-1}) \frac{\partial K_y}{\partial \sigma_f^2} = \begin{bmatrix} -1,2882 & 0,4662 \\ 0,4662 & 2,2743 \end{bmatrix} \begin{bmatrix} 1 & 0,6065 \\ 0,6065 & 1 \end{bmatrix} = \begin{bmatrix} -1,0054 & -0,3151 \\ 1,8456 & 2,5570 \end{bmatrix} \quad (12)$$

$$\text{tr} \left( (\alpha\alpha^T - K_y^{-1}) \frac{\partial K_y}{\partial \sigma_f^2} \right) = -1,0054 + 2,5570 = 1,5516 \quad (13)$$

$$\frac{\partial \log p(\mathbf{y}|X, \boldsymbol{\theta})}{\partial \sigma_f^2} = \frac{1}{2}(1,5516) = 0,7758 \implies -\frac{\partial \log p(\mathbf{y}|X, \boldsymbol{\theta})}{\partial \sigma_f^2} = -0,7758 \quad (14)$$

b. Terhadap  $\ell$ :

$$\frac{\partial K_y}{\partial \ell} = \begin{bmatrix} 0 & 0,6065 \cdot \frac{r_{12}^2}{1^3} \\ 0,6065 \cdot \frac{r_{12}^2}{1^3} & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0,6065 \\ 0,6065 & 0 \end{bmatrix} \quad (15)$$

$$(\alpha\alpha^T - K_y^{-1}) \frac{\partial K_y}{\partial \ell} = \begin{bmatrix} -1,2882 & 0,4662 \\ 0,4662 & 2,2743 \end{bmatrix} \begin{bmatrix} 0 & 0,6065 \\ 0,6065 & 0 \end{bmatrix} = \begin{bmatrix} 0,2828 & -0,7813 \\ 1,3794 & 0,2828 \end{bmatrix} \quad (16)$$

$$\text{tr} \left( (\alpha\alpha^T - K_y^{-1}) \frac{\partial K_y}{\partial \ell} \right) = 0,2828 + 0,2828 = 0,5656 \quad (17)$$

$$\frac{\partial \log p(\mathbf{y}|X, \boldsymbol{\theta})}{\partial \ell} = \frac{1}{2}(0,5656) = 0,2828 \implies -\frac{\partial \log p(\mathbf{y}|X, \boldsymbol{\theta})}{\partial \ell} = -0,2828 \quad (18)$$

c. Terhadap  $\sigma_n^2$ :

$$\frac{\partial K_y}{\partial \sigma_n^2} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2 \quad (19)$$

$$(\alpha\alpha^T - K_y^{-1}) \frac{\partial K_y}{\partial \sigma_n^2} = \begin{bmatrix} -1,2882 & 0,4662 \\ 0,4662 & 2,2743 \end{bmatrix} \quad (20)$$

$$\text{tr} \left( (\alpha\alpha^T - K_y^{-1}) \frac{\partial K_y}{\partial \sigma_n^2} \right) = -1,2882 + 2,2743 = 0,9861 \quad (21)$$

$$\frac{\partial \log p(\mathbf{y}|X, \boldsymbol{\theta})}{\partial \sigma_n^2} = \frac{1}{2}(0,9861) = 0,4930 \implies -\frac{\partial \log p(\mathbf{y}|X, \boldsymbol{\theta})}{\partial \sigma_n^2} = -0,4930 \quad (22)$$

## 6. Update Parameter

$$\nabla_{\boldsymbol{\theta}} (-\log p(\mathbf{y}|X, \boldsymbol{\theta})) = \begin{bmatrix} -\frac{\partial \log p(\mathbf{y}|X, \boldsymbol{\theta})}{\partial \sigma_f^2} \\ -\frac{\partial \log p(\mathbf{y}|X, \boldsymbol{\theta})}{\partial \ell} \\ -\frac{\partial \log p(\mathbf{y}|X, \boldsymbol{\theta})}{\partial \sigma_n^2} \end{bmatrix} = \begin{bmatrix} -0,7758 \\ -0,2828 \\ -0,4930 \end{bmatrix} \quad (23)$$

$$\boldsymbol{\theta}^{(1)} = \boldsymbol{\theta}^{(0)} - \eta \nabla_{\boldsymbol{\theta}} (-\log p(\mathbf{y}|X, \boldsymbol{\theta})) \quad (24)$$

$$\begin{bmatrix} (\sigma_f^2)^{(1)} \\ \ell^{(1)} \\ (\sigma_n^2)^{(1)} \end{bmatrix} = \begin{bmatrix} 1,0000 \\ 1,0000 \\ 0,1000 \end{bmatrix} - 0,1 \begin{bmatrix} -0,7758 \\ -0,2828 \\ -0,4930 \end{bmatrix} = \begin{bmatrix} 1,0776 \\ 1,0283 \\ 0,1493 \end{bmatrix} \quad (25)$$

## 1.2 Adam

Inisialisasi:  $\eta = 0,1$ ,  $\beta_1 = 0,9$ ,  $\beta_2 = 0,999$ ,  $\epsilon = 10^{-8}$ , serta  $\mathbf{m}^{(0)} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ ,  $\mathbf{v}^{(0)} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ .

Menggunakan gradien dari perhitungan sebelumnya:

$$-\nabla_{\boldsymbol{\theta}} (\log p(\mathbf{y}|X, \boldsymbol{\theta})) = \begin{bmatrix} -0,7758 \\ -0,2828 \\ -0,4930 \end{bmatrix}$$

### 1. Estimasi Momen Pertama & Kedua

$$\begin{aligned} \mathbf{m}^{(1)} &= \beta_1 \mathbf{m}^{(0)} + (1 - \beta_1) (-\nabla_{\boldsymbol{\theta}} \log p(\mathbf{y}|X, \boldsymbol{\theta})) \\ &= 0,9 \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + 0,1 \begin{bmatrix} -0,7758 \\ -0,2828 \\ -0,4930 \end{bmatrix} = \begin{bmatrix} -0,0776 \\ -0,0283 \\ -0,0493 \end{bmatrix} \end{aligned} \quad (26)$$

$$\begin{aligned} \mathbf{v}^{(1)} &= \beta_2 \mathbf{v}^{(0)} + (1 - \beta_2) (-\nabla_{\boldsymbol{\theta}} \log p(\mathbf{y}|X, \boldsymbol{\theta}))^2 \\ &= 0,999 \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + 0,001 \begin{bmatrix} (-0,7758)^2 \\ (-0,2828)^2 \\ (-0,4930)^2 \end{bmatrix} = \begin{bmatrix} 0,000602 \\ 0,000080 \\ 0,000243 \end{bmatrix} \end{aligned} \quad (27)$$

### 2. Koreksi Bias

$$\hat{\mathbf{m}}^{(1)} = \frac{\mathbf{m}^{(1)}}{1 - \beta_1^1} = \frac{1}{0,1} \begin{bmatrix} -0,0776 \\ -0,0283 \\ -0,0493 \end{bmatrix} = \begin{bmatrix} -0,7758 \\ -0,2828 \\ -0,4930 \end{bmatrix} \quad (28)$$

$$\hat{\mathbf{v}}^{(1)} = \frac{\mathbf{v}^{(1)}}{1 - \beta_2^1} = \frac{1}{0,001} \begin{bmatrix} 0,000602 \\ 0,000080 \\ 0,000243 \end{bmatrix} = \begin{bmatrix} 0,6019 \\ 0,0800 \\ 0,2430 \end{bmatrix} \quad (29)$$

### 3. Update Parameter

$$\begin{aligned} \boldsymbol{\theta}^{(1)} &= \boldsymbol{\theta}^{(0)} - \eta \frac{\hat{\mathbf{m}}^{(1)}}{\sqrt{\hat{\mathbf{v}}^{(1)} + \epsilon}} \\ \begin{bmatrix} (\sigma_f^2)^{(1)} \\ \ell^{(1)} \\ (\sigma_n^2)^{(1)} \end{bmatrix} &= \begin{bmatrix} 1,0000 \\ 1,0000 \\ 0,1000 \end{bmatrix} - 0,1 \begin{bmatrix} \frac{-0,7758}{\sqrt{0,6019+10^{-8}}} \\ \frac{-0,2828}{\sqrt{0,0800+10^{-8}}} \\ \frac{-0,4930}{\sqrt{0,2430+10^{-8}}} \end{bmatrix} = \begin{bmatrix} 1,0000 \\ 1,0000 \\ 0,1000 \end{bmatrix} - 0,1 \begin{bmatrix} -1,0000 \\ -1,0000 \\ -1,0000 \end{bmatrix} = \begin{bmatrix} 1,1000 \\ 1,1000 \\ 0,2000 \end{bmatrix} \end{aligned} \quad (30)$$

## 2 Eksplorasi Operasi Kombinasi Kernel

Empat kombinasi kernel yang diuji pada eksperimen ini didefinisikan sebagai:

$$k_{\text{RBF}+\text{Per}}(x, x') = \sigma_f^2 \exp\left(-\frac{(x-x')^2}{2\ell_{\text{RBF}}^2}\right) + \sigma_p^2 \exp\left(-\frac{2 \sin^2\left(\frac{\pi|x-x'|}{p}\right)}{\ell_{\text{Per}}^2}\right) \quad (31)$$

$$k_{\text{RBF} \times \text{Per}}(x, x') = \sigma_f^2 \sigma_p^2 \exp\left(-\frac{(x-x')^2}{2\ell_{\text{RBF}}^2} - \frac{2 \sin^2\left(\frac{\pi|x-x'|}{p}\right)}{\ell_{\text{Per}}^2}\right) \quad (32)$$

$$k_{\text{Lin}+\text{Per}}(x, x') = \sigma_b^2 + \sigma_v^2(x-c)(x'-c) + \sigma_p^2 \exp\left(-\frac{2 \sin^2\left(\frac{\pi|x-x'|}{p}\right)}{\ell_{\text{Per}}^2}\right) \quad (33)$$

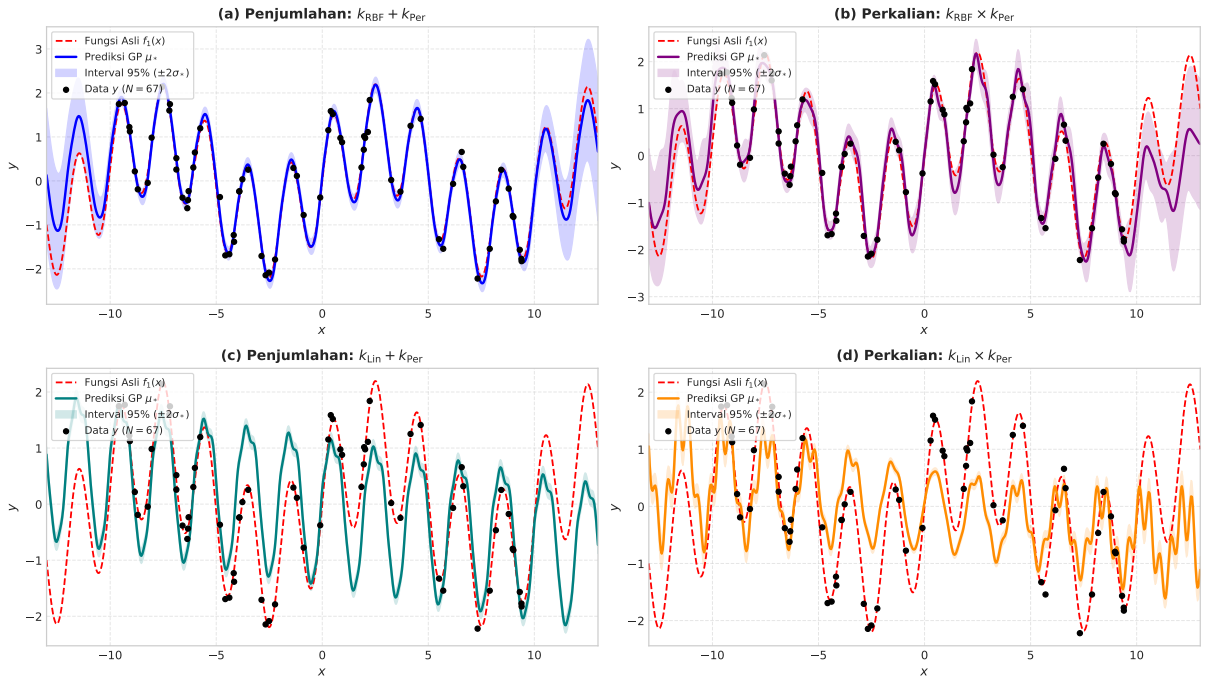
$$k_{\text{Lin} \times \text{Per}}(x, x') = [\sigma_b^2 + \sigma_v^2(x-c)(x'-c)] \times \sigma_p^2 \exp\left(-\frac{2 \sin^2\left(\frac{\pi|x-x'|}{p}\right)}{\ell_{\text{Per}}^2}\right) \quad (34)$$

### 2.1 Eksperimen 1

Fungsi asli yang digunakan:

$$f_1(x) = \sin(0,6x) + 1,2 \sin\left(\frac{2\pi x}{2,0}\right) \quad (35)$$

dengan  $N = 67$  titik data observasi  $y_i = f_1(x_i) + \varepsilon_i$ ,  $\varepsilon_i \sim \mathcal{N}(0, 0,15^2)$  pada rentang  $x \in [-10, 10]$ .



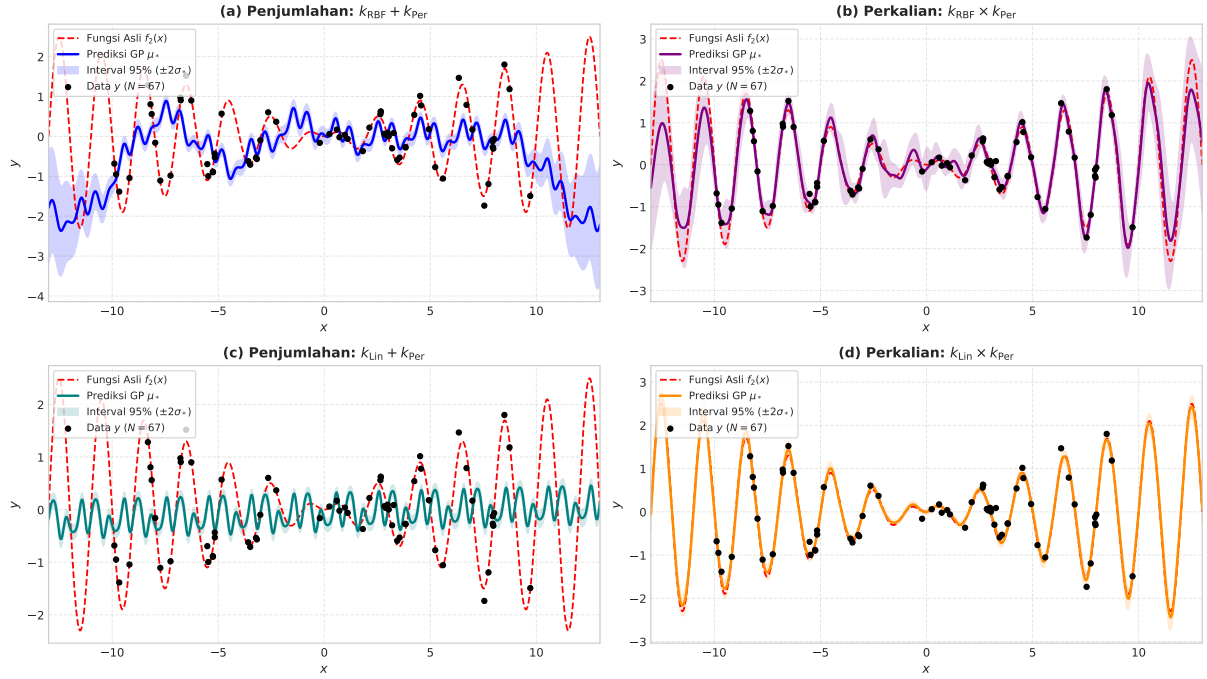
Gambar 2: Hasil *fitting* Gaussian Process Regression pada fungsi asli  $f_1(x)$  (35): (a)  $k_{\text{RBF}} + k_{\text{Per}}$ ; (b)  $k_{\text{RBF}} \times k_{\text{Per}}$ ; (c)  $k_{\text{Lin}} + k_{\text{Per}}$ ; (d)  $k_{\text{Lin}} \times k_{\text{Per}}$ .

### 2.2 Eksperimen 2

Fungsi asli yang digunakan:

$$f_2(x) = 0,2x \sin\left(\frac{2\pi x}{2,0}\right) \quad (36)$$

dengan  $N = 67$  titik data observasi  $y_i = f_2(x_i) + \varepsilon_i$ ,  $\varepsilon_i \sim \mathcal{N}(0, 0.15^2)$  pada rentang  $x \in [-10, 10]$ .



Gambar 3: Hasil *fitting* Gaussian Process Regression pada fungsi asli  $f_2(x)$  (36): (a)  $k_{\text{RBF}} + k_{\text{Per}}$ ; (b)  $k_{\text{RBF}} \times k_{\text{Per}}$ ; (c)  $k_{\text{Lin}} + k_{\text{Per}}$ ; (d)  $k_{\text{Lin}} \times k_{\text{Per}}$ .

## Pustaka

- [1] Rasmussen, C. E., & Williams, C. K. (2006). *Gaussian Processes for Machine Learning*. MIT Press, Cambridge, MA.